Elimination of the GGHV Branch-(c) Normal Form at Degree Pair $(72, 108)$:

A Certified Computational Audit

Descent, Chart Conditions, and the Rank-Lemma Obstruction

Brandon D. McCrary

Independent Researcher

ORCID: 0009-0008-2804-7165

Computation, Lean and LaTeX Source:
https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace

October 1, 2026

Abstract

We present a computational elimination of the branch-(c) normal form arising from case (1) of Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical $(72, 108)$ counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted. This paper is the branch-(c) companion to the branch-(a,b) elimination [2], whose format it follows.

Starting from the verified Newton polygons

$$N(P) = \mathrm{conv}\{(0,0),(1,0),(8,14),(8,16),(0,8)\},$$
$$N(Q) = \mathrm{conv}\{(0,0),(2,1),(12,21),(12,24),(0,12)\},$$

with $[P, Q] = \lambda x^2$ post-φ and $\lambda \neq 0$, we reduce the bracket condition to layer identities E_n in the grading $w(x) = 2$, $w(y) = -1$, extract 132 exact scalar equations, and eliminate the layers $E_4, \dots, E_{-2}$ to twelve chart conditions $\Omega, \Psi, \Phi_1, \Phi_2, \Theta_1, \Theta_2, \Theta_3$ plus five pure rows, over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$. The top layer is the branch-(a,b) problem, whose exact K_5 classification is inherited. The descent is proved twice: symbolically, as 600 kernel-checked reflective polynomial identities in Lean 4 (no `sorry`, standard axioms only), and numerically, by exact finite-field linear algebra.

The chart is empty. On the stratum $b_{11,20} = 0$ this is proved in the Lean kernel from an exact K_5 certificate (degree 2, 75 terms). On the whole chart it follows from a rank lemma: full row rank of the fixed-degree Macaulay matrix lifts from $\mathbb{F}_p$ to K_5 through a nonzero minor. At weight $W = 24$ the 3199×6054 Macaulay matrix has full row rank 3199 at two primes ($p = 1000003$, $p = 32003$), verified by two independent implementations (FLINT and NumPy), with negative controls. An earlier reduction lemma used in a preliminary version of this work is false as stated and is retracted here; the rank lemma is its correct replacement, and the distinction—rank lifts, mere consistency does not—is proved by counterexample.

In Lean, with all 790 modules built and `#print axioms` reporting only `propext`, `Classical.choice`, `Quot.sound`,

$$\texttt{ChartEmptyC_T1ne0}\ L \Rightarrow \neg\exists\, P\, Q\, \lambda,\ \lambda \neq 0 \wedge \texttt{NewtonNFc}\ P\, Q \wedge [P, Q] = \lambda x^2,$$

and the rank computation supplies `ChartEmptyC_T1ne0` outside Lean, at epistemic grade B. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.4).

Contents

1 Introduction **4**

1.1 Context and prior work 4

1.2 Main result 5

1.3 Logical status 5

1.4 Outline 6

2 Source data **6**

3 The polynomial system **7**

3.1 Lattice points 7

3.2 The bracket equations 7

4 Exact layer reduction **7**

5 Normalization **8**

6 The top layer **8**

7 Descent **8**

8 The chart conditions **9**

8.1 Chart variables 9

8.2 The twelve conditions 10

8.3 The Ω parabola 10

9 The stratum $b_{11,20} = 0$ **10**

10 The whole chart **12**

10.1 The rank lemma 12

10.2 The instance 13

11 Proof of the main theorem 14

12 Epistemic status and what is not claimed 14

13 Conclusion 15

14 Prior work, attribution, and independence 15

A Computational archive 16

B Lean module map 17

C Computational and AI-assisted research disclosure 17

1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [5]) to 108, leaving open only the degree pair $(72, 108)$ and its symmetric $(108, 72)$.

For the degree pair $(72, 108)$, [1] (an unrefereed preprint, arXiv:2204.14178v1) identifies two cases. The corner $(9, 27)$ is discarded in [1, §5]; the open case is the corner $(8, 28)$ with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ, to one of two Newton polygon configurations. Case (2), branches (a) and (b), is eliminated in [2]. Case (1), the subject of this paper, is branch (c):

$$\begin{aligned} N(P) &= \operatorname{conv}\{(0,0), (1,0), (8,14), (8,16), (0,8)\}, \\ N(Q) &= \operatorname{conv}\{(0,0), (2,1), (12,21), (12,24), (0,12)\}. \end{aligned} \tag{1}$$

with $[P, Q] = \lambda x^2$ in post-φ coordinates and $\lambda \neq 0$. The additional vertices $(0, 8)$ and $(0, 12)$ distinguish branch (c) from branches (a,b). See Figure 1 for the polygon shapes.

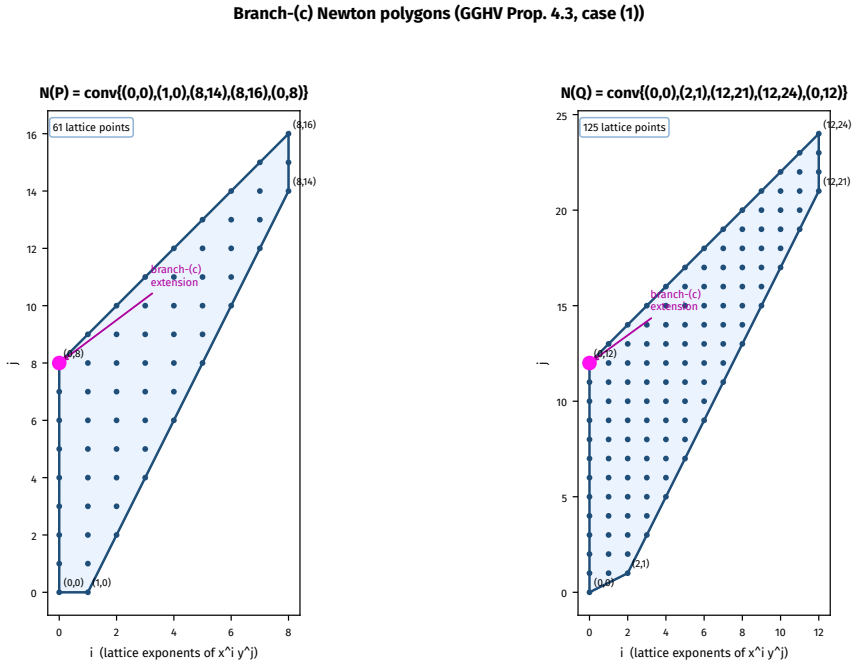


Figure 1: Branch-(c) Newton polygons, rendered from the exact vertex data of (1) (source-gated at GGHV v1 lines 495 and 599–600). In each panel the branch-(a,b) sub-polygon is drawn in the lighter shade underneath; the extra branch-(c) vertices $(0, 8)$ and $(0, 12)$ are highlighted in the accent color. All lattice points are marked (61 for $N(P)$, 125 for $N(Q)$); the ten vertex coefficients required nonzero by `NewtonNFc` are labelled. Rendered by `figures/fig1_newton_polygons.py` from the vertex lists.

Santibáñez-Leal has published machine evidence related to the $(72, 108)$ candidate [6, 7], explicitly not claiming elimination. The candidate itself is documented in [8]. Independent non-peer-reviewed treatments of the GGHV configuration in 2026 include Helali's exact characteristic-zero certificate [9] and the structural analysis of wstrinz [10]; none is peer-

reviewed, and the parity-constrained Davenport–Stothers equation relevant to the top layer remains unclassified.

1.2 Main result

Definition 1.1 (Branch-(c) Newton normal form)**.** Let L be a field of characteristic 0 and $P, Q \in L[x, y]$. `NewtonNFc`(P, Q) holds when $\operatorname{supp} P \subset N(P)$ and $\operatorname{supp} Q \subset N(Q)$ with $N(P), N(Q)$ as in (1), and the ten vertex coefficients (at $(0,0), (1,0), (8,14), (8,16), (0,8)$ of P and $(0,0), (2,1), (12,21), (12,24), (0,12)$ of Q) are all nonzero. In Lean this is `NewtonNFc` (`Jacobian/BranchC/DescentClaim.lean`); see listing 1.

Theorem 1.2 (Elimination of the GGHV branch-(c) normal form, conditional)**.** *Let L be a field of characteristic 0 satisfying* ***`ChartEmptyC_T1ne0`*** *(listing 3): the twelve chart conditions of §8 have no common zero with $b_{12,22} = 1$ on the stratum $b_{11,20} \neq 0$. Then there are no $P, Q \in L[x, y]$ and no $\lambda \neq 0$ with* ***`NewtonNFc`***(P, Q) *and* $[P, Q] = \lambda x^2$.

The hypothesis `ChartEmptyC_T1ne0` is a precise Lean `Prop`. It is established outside Lean in §10, at epistemic grade B (see §12).

Corollary 1.3 (Unconditional elimination under computational premises)**.** *Under the computational premises of §10 (the rank lemma, Lemma 10.1, instantiated at two primes by two independent implementations, with negative controls),* ***`ChartEmptyC_T1ne0`*** *holds for every field L of characteristic 0, and hence no P, Q, λ as in Theorem 1.2 exist.*

Corollary 1.4 (Jacobian application)**.** *Assume [1, Proposition 4.3] is correct and exhaustive. Then no counterexample to the planar Jacobian conjecture of degree pair $(72, 108)$ arises from branch (c).*

1.3 Logical status

Remark 1.5. Theorem 1.2 does not depend on GGHV: its Lean proof goes from `NewtonNFc` and $[P, Q] = \lambda x^2$ through the descent to the chart statement, all inside Lean. In Lean, `main_theorem_c_of_chartEmpty_T1ne0` (listing 4) derives the conclusion from the explicit hypothesis `ChartEmptyC_T1ne0` (listing 3), with `#print axioms` reporting only `propext`, `Classical.choice` and `Quot.sound`, and no `sorry`, `axiom` or `native_decide` anywhere in the 790 modules. The chain is: `NewtonNFc` $+ [P, Q] = \lambda x^2 \xrightarrow{\text{B0–B3}}$ `DescentClaimC` hypotheses $\xrightarrow{\text{B4–B5}}$ `ChartEmptyC` $\xrightarrow{\text{B6–B7}}$ `ChartEmptyC_T1ne0` $\Rightarrow$ contradiction. Only Corollary 1.4 additionally assumes GGHV Proposition 4.3.

Remark 1.6 (A retraction). A preliminary version of this work claimed a "reduction lemma": that a unit ideal modulo a prime lifts to a unit ideal over $\mathbb{Q}$. That statement is **false**. Counterexample: $F = 1 + 109X$ is 1 modulo 109 (unit ideal) but vanishes at $X = -1/109$ over $\mathbb{Q}$ (proper ideal). The Lean file deriving `False` from the stated axiom is archived for the record, and every theorem depending on it is vacuous. The correct replacement is the rank lemma (Lemma 10.1): *full row rank* of a fixed-degree Macaulay matrix lifts through a nonzero minor, because nonvanishing of a determinant is an open condition. Mere consistency of a modular system does not lift. §12 gives the precise distinction.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the polynomial system and the grading. Section 4 gives the exact layer reduction and the 132 raw equations. Section 5 fixes normalizations and reaches the chart. Section 6 treats the top layer, inherited from branches (a,b). Section 7 carries out the descent to the twelve conditions, proved by kernel reflection. Section 8 studies the chart conditions. Section 9 eliminates the stratum $b_{11,20} = 0$ inside Lean. Section 10 eliminates the whole chart by the rank lemma. Section 11 assembles the proof of Theorem 1.2. Section 12 gives the epistemic accounting: grades, the retraction, and what is not claimed. Section 13 concludes. Section 14 covers prior work, attribution, and independence. Figure 2 shows the bridge architecture.

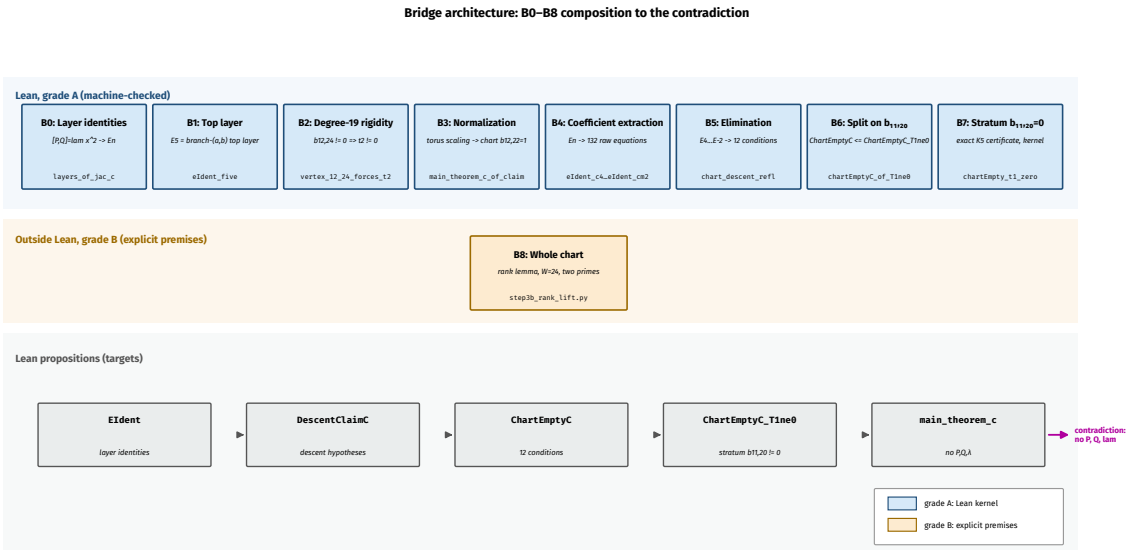


Figure 2: Bridge architecture. Top band: the eight Lean bridges B0–B7 (grade A), each labelled with its certified Lean name. Middle band: the external rank lemma B8 (grade B), the only non-Lean component. Bottom band: the Lean propositions they target, composing left to right to the final contradiction. Grade colors are consistent with Table 11. Rendered by `figures/fig2_bridges.py`.

2 Source data

Every base object below is source-gated directly against GGHV v1 before any proof is drafted, following the discipline of [2]. The branch-(c) Newton polygons (1) are [1, Proposition 4.3, case (1)]: the vertex lists are asserted at GGHV v1 lines 495 and 599–600, and the checker `source_gate_verify_c.py` re-asserts them byte-for-byte against the downloaded v1 text. The post-φ bracket target $[P, Q] = \lambda x^2$ is [1, §5].

*Listing 1: The branch-(c) Newton normal form (**`NewtonNFc`**)*

```
-- Jacobian/BranchC/DescentClaim.lean (excerpt, exact statement)
def NewtonNFc (P Q : MvPolynomial (Fin 2) L) : Prop :=
  (P.support \subseteq NPc) \land (Q.support \subseteq NQc) \land
  TenVertexCoeffsNonzero P Q
```

```
-- NPc = conv {(0,0),(1,0),(8,14),(8,16),(0,8)}
-- NQc = conv {(0,0),(2,1),(12,21),(12,24),(0,12)}
```

The Lean formalization works over an arbitrary field L of characteristic 0. The quintic field

$$K_5 = \mathbb{Q}[w]/(R), \qquad R(w) = w^5 - w^4 + 3w^3 + 3w^2 + 26, \tag{2}$$

enters through the top layer (§6): R is irreducible over $\mathbb{Q}$, and it remains irreducible modulo 109, so $\mathbb{F}_{109^5} = \mathbb{F}_{109}[w]/(R)$ is a field. The good prime/root pairs used in this audit are

$$(101, 9), \qquad (1000003, 806739), \qquad (32003, 11147), \tag{3}$$

each satisfying $R(w_0) \equiv 0 \pmod{p}$, plus the inert prime 109.

3 The polynomial system

3.1 Lattice points

Write $P = \sum a_{i,j} x^i y^j$, $Q = \sum b_{i,j} x^i y^j$. The branch-(c) supports are the lattice points of the polygons (1); the supports strictly contain those of branches (a,b) because of the vertices $(0, 8)$ and $(0, 12)$. The Lean files record the supports as `rangeP`, `rangeQ` (`Jacobian/BranchC/DescentClaim.lean`).

3.2 The bracket equations

Grade by $w(x) = 2$, $w(y) = -1$. Write the layers

$$P = \sum_k y^{-k} A_k, \qquad Q = \sum_l y^{-l} B_l, \qquad A_k, B_l \in L[u]$$

with u the residual variable. The bracket of weighted-homogeneous pieces is weighted-homogeneous, so $[P, Q] = \lambda x^2$ splits into one identity per weight:

$$\texttt{EIdent}\ A\ B\ \lambda\ n: \quad \sum_{k+l=n} \mathrm{LT}_{k,l}(A_k, B_l) = [n = 5] \cdot \lambda u^2, \qquad \mathrm{LT}_{k,l}(A, B) = kAB' - lA'B. \tag{4}$$

Layer k of P corresponds to $W = j - 2i$ via $n = 1 - W$; the raw equation key (i, j) corresponds to $(n = 2i + 1 - j,\ m = i)$, and the coefficient of the product term is $c = k \cdot (m + 1 - i_1) - l \cdot i_1$ with $l = n - k$ (§5.3 of the v3.1 guide). The top layer $n = 5$ carries the λu^2 term; all other layers are homogeneous.

4 Exact layer reduction

B0 (Lean, v1). From `jac`$(P, Q) = C\,\lambda \cdot x^2$ and `NewtonNFc`, the layer identities `EIdent` $A\ B\ \lambda\ n$ hold for every layer $n \geq -20$ (`layers_of_jac_c`, `layer_bracket_gen` in `LayersGen.lean`). This is an exact symbolic identity, verified by the Lean kernel.

B4 (Lean, v3). Coefficient extraction turns the polynomial identities `EIdent` $A\ B\ 0\ n$, $n = 4, \dots, -2$, into scalar equations. For each monomial of $\mathrm{LT}_{k,l}(A_k, B_l)$, `coeff_layerTermZ` computes its coefficient with integer indices; coefficients outside the branch-(c) supports vanish by `rangeP`, `rangeQ` (179 zero facts; $a_{0,0}, b_{0,0}$ are in the support but always carry coefficient 0). Each of the 132 raw equations then follows by `linear_combination`. This is the recipe of the repository's `BranchAbFinal.lean`, replayed by `certgen_c/gen_bridge_c.py` (`Descent2R/Bridge.lean`).

5 Normalization

B1 (Lean, v1). The top layer $n = 5$ is the branch-(a,b) problem, classified exactly over K_5 in [2] and formalized in the repository (`ChartProof/Final.lean`, `BranchAb*`). Branch (c) inherits its exact K_5 solution; this is where w and R enter (`eIdent_five`).

B2 (Lean, v1/v2). Degree-19 rigidity: $b_{12,24} \neq 0$ forces $t_2 = b_{12,22} \neq 0$. The theorems are `vertex_12_24_forces_t2` and `edge19_rigidity_of_jac` in `Edge19.lean`, and `t0_vertex_rigidity_of_jac` in `LayerE2.lean`, `Degree19.lean`).

B3 (Lean, v1). Torus and weighted scaling remove the continuous symmetries, normalizing to the rescaled K_5 top layer and the chart $b_{12,22} = 1$ (`eIdent_transport`, `main_theorem_c_of_claim` in `DescentClaim.lean`). These are `DescentClaimC`'s hypotheses: the 19 top-layer values, the 123 unknowns, and the 132 raw equations.

6 The top layer

At $n = 5$ the branch-(c) layer equations coincide with the branch-(a,b) top layer, whose complete K_5 classification is proved in [2] (five torus orbits; the degree-35 eliminant irreducible over $\mathbb{Q}$). The classification is used here as a black box: the 19 top-layer values enter `Descent2R/Main.lean` as hypotheses `h_v` at minimal integer scale, and the K_5 arithmetic they satisfy is exactly the arithmetic of (2). No new top-layer analysis is needed for branch (c); the new mathematics is the descent of §7 and the chart analysis of §§8–10.

Formalization seam. Julian's 2026-09-26 audit identified one genuine seam at this step: the degree-19 rigidity lemma is *not* formalized inside Lean. The parity-constrained Davenport–Stothers classification (the step ruling out $\deg f > 19$) is supplied by the external certified point in `e5_cert.json` (exact equality over K_5), which Lean imports and checks by `decide`; the mathematical classification argument itself is grade B, not grade A. This is the only non-kernel step below the chart (the other gap flagged by the audit, the missing GGHV citation line, is bibliographic and is repaired in §14).

7 Descent

B5 (Lean, v3; kernel reflection). Each layer $n = 4, \dots, -2$ is a linear system in its new unknowns, with coefficients polynomial in the earlier unknowns and the chart parameters. The v2 operator-rank theorems (`Rank/**`, 96 modules) give each system's exact rank over K_5; the left null vectors give the elimination conditions:

Layer	Modules	Conditions produced
E_4 ($n = 4$)	`Descent2R/L4` (54)	—
E_3 ($n = 3$)	`L3` (57)	—
E_2 ($n = 2$)	`L2c` (57)	none (left null vector annihilates RHS)
E_1 ($n = 1$)	`L1` (60)	Ω
E_0 ($n = 0$)	`L0` (60)	Ψ and pure row `E0pure_18_37`
E_{-1} ($n = -1$)	`Lm1` (99)	Φ_1, Φ_2 and pure rows `Em1pure_17_36`, `Em1pure_18_38`
E_{-2} ($n = -2$)	`Lm2` (110)	$\Theta_1, \Theta_2, \Theta_3$ and pure rows `Em2pure_16_35`, `Em2pure_17_37`

`lower_c.py` computed these eliminations in Python; `Descent2R` replays them as 600 polynomial identities target $= \sum m_i \cdot \text{fact}_i$, each checked by the Lean kernel. Reflection uses the repository's `Jacobian/ChartProof/Reflect.lean` (built on core's `Lean.Grind.CommRing`): `Expr`/`Poly`, normalized by `toPolyK` and compared by `decide +kernel`. Each step theorem has the form `(Expr.mul (.num K) X).denote ctx = 0` proved by `lc_zero` with a kernel `decide`; `Main` removes K with `cancel_int` when $K \neq 1$ (231 of 752 `have`s). The context is 143 atoms in a balanced `RArray` (`gctx2`). A tactic proof of the same identities was tried first and abandoned: 214 MB of source, chunks over 10 min each. The kernel evaluates the normalizer directly with GMP arithmetic; `decide +kernel` is not `native_decide`—no compiler is trusted.

The main theorem of the descent (`Descent2R/Main.lean`):

Listing 2: ***`chart_descent_refl`****: the twelve conditions from the 132 raw equations*

```
-- Descent2R/Main.lean (statement; 752 haves elided)
theorem chart_descent_refl
  (w : L) (hw : R w = 0)
  (v : Fin 19 \to L) (h_v : v = K5_top_layer_values)
  (u : Fin 123 \to L)
  (h4_ : raw_equations_n4) ... (hm2_ : raw_equations_nm2) :
  cond_Omega w ... = 0 \land cond_Psi w ... = 0 \land
  cond_Phi1 w ... = 0 \land cond_Phi2 w ... = 0 \land
  cond_Theta1 w ... = 0 \land cond_Theta2 w ... = 0 \land cond_Theta3 w ... = 0 \land
  cond_E0pure_18_37 w ... = 0 \land cond_Em1pure_17_36 w ... = 0 \land
  cond_Em1pure_18_38 w ... = 0 \land cond_Em2pure_16_35 w ... = 0 \land
  cond_Em2pure_17_37 w ... = 0
```

The twelve conjuncts are `lower_c`'s conditions, defined once in `CondsC.lean` (generated by `certgen_c/gen_conds_c.py`): each `cond_*` is the explicit polynomial with the same balanced sum tree as the reflected fact (subtrees of at most 32 terms are separate definitions: 110 definitions for 12 conditions), so `ek.denote ctx = 0` and `cond_* … = 0` are definitionally equal. `Bridge` (`Descent2R/Bridge.lean`) then gives `descentClaimC_of_chartEmpty`: `DescentClaimC` ⇐ `ChartEmptyC`.

8 The chart conditions

8.1 Chart variables

On the normalized chart $b_{12,22} = 1$ the conditions involve seven parameters:

Symbol	T_1	T_2	S_1	S_2	R_1	R_2	Q
Coefficient	$b_{11,20}$	$b_{12,22}$	$b_{11,21}$	$b_{12,23}$	$a_{6,13}$	$a_{7,15}$	$b_{10,21}$
Weight	1	1	2	2	3	3	4

8.2 The twelve conditions

`certgen_c/conds_c.json` (md5 `168299d255af361578423c0ed4852c11`, written by `certgen_c/lower_c.py`):

Condition	K_5-terms	Integer terms (minimal scale, w expanded)
Ω	3	15
Ψ	32	160
Φ_1, Φ_2	54, 54	270, 270
Θ_1, Θ_2, Θ_3	84, 84, 84	420, 420, 420
`E0pure_18_37`	9	41
`Em1pure_17_36`	39	195
`Em1pure_18_38`	12	60
`Em2pure_16_35`	77	381
`Em2pure_17_37`	55	275

8.3 The Ω parabola

Exactly over K_5,

$$\Omega = o_1 S_2^2 + o_2 T_2^2 S_2 + o_3 T_2^4, \qquad o_2^2 = 4 o_1 o_3, \quad o_1 \neq 0, \tag{5}$$

so that with $\kappa = -o_2/(2o_1) \in K_5$,

$$\Omega = o_1 (S_2 - \kappa T_2^2)^2. \tag{6}$$

The relation $o_2^2 = 4 o_1 o_3$ is checked in `step1/lowerc_chart.py` and independently in `step3c_chart_identity.py`; $o_1(w) \neq 0$ is proved in the Lean kernel by a Bezout identity $a \cdot N_{o_1} + b \cdot R = c$ with c a 151-digit constant (`T1Zero.s_X2`). On the chart $T_2 = 1$, $\Omega = 0$ forces $S_2 = \kappa$: the chart collapses to the parabola's vertex line (Figure 3).

9 The stratum $b_{11,20} = 0$

B7 (Lean kernel). On the stratum $T_1 = b_{11,20} = 0$ the conditions admit an exact K_5 certificate small enough for the kernel: degree 2, 75 K_5-terms, coefficient heights at most 9,681 digits (`chart_certificates/step1/certB_lowerc.json`, produced by `certB_lowerc.py` via pivots from a mod-p RREF and an exact FLINT solve; independently rechecked by `check_certB_lowerc.py` with `fmpq_mpoly` expansion, w as a variable, reduced mod R at the end). `T1Zero` checks the certificate chain end to end in the kernel (`gen_t1zero_lean.py`):

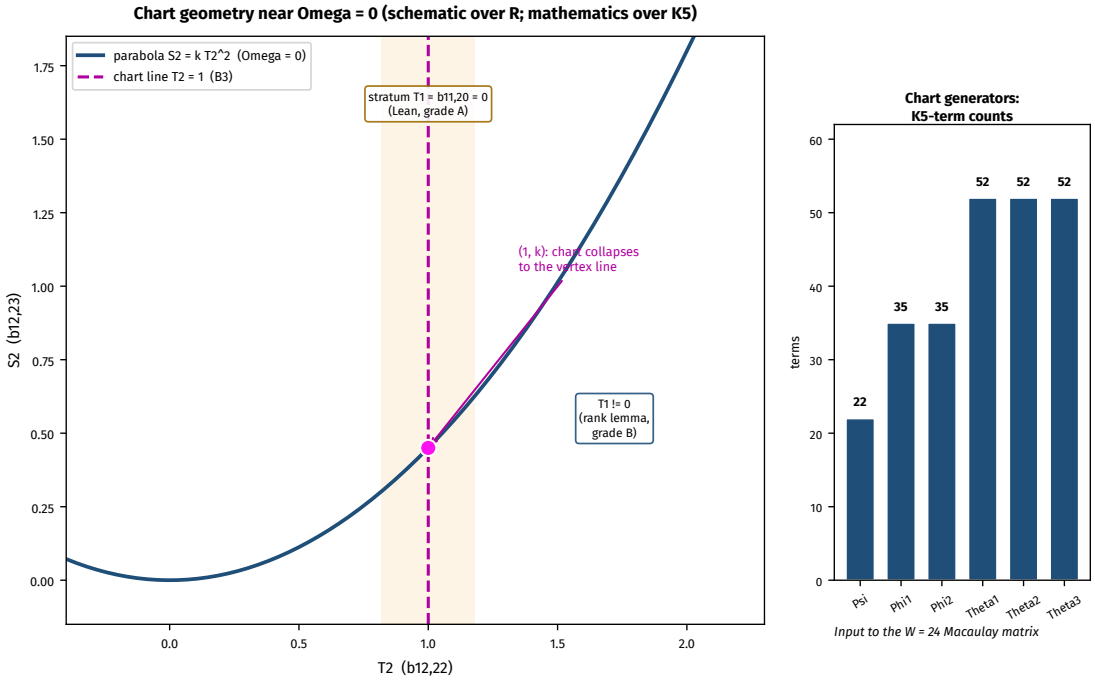


Figure 3: Chart geometry near $\Omega = 0$. Main panel: the (T_2, S_2)-plane drawn schematically over $\mathbb{R}$ for illustration (the mathematics is over K_5), showing the parabola $S_2 = \kappa T_2^2$ from (6), the chart line $T_2 = 1$ (normalization B3), and their intersection $(1, \kappa)$; the stratum $T_1 = b_{11,20} = 0$ is indicated as a transverse slice, with the $T_1 = 0$ stratum (Lean, grade A) and the $T_1 \neq 0$ stratum (rank lemma, grade B) labelled. Inset: the six chart generators' K_5-term counts $(22, 35, 35, 52, 52, 52)$, the input to the Macaulay matrix of §10. Rendered by `figures/fig3_omega.py`.

Step	Identity (integer-polynomial, kernel-checked)	Build
s_sq	$D_O D_{o_1} D_k^2 \Omega \equiv D_O N_{o_1} X^2 \bmod (T_2 - 1, R), X = D_k S_2 - N_\kappa(w), D_k = 851565312$	1.5 s
s_X2	Bezout $a \cdot e_{sq} + b \cdot D_O X^2 R = c \cdot D_O X^2$ (c: 151 digits) $\Rightarrow X = 0$	—
s_f_n (6)	$D_k^d \cdot ck_n = D_k^d T_1 A_1 + D_k^d (T_2 - 1) A_2 + X A_3 + f_n + R A_4$	2.4–5.7 s
s_p_n (6)	$p_n = v_n f_n - q_n R$ (250–375 terms, ≈ 5,440 digits)	17–28 s
s_final	$\sum_n p_n = DD$, a 5387-digit integer, exact	55.6 s

chartEmpty_t1_zero chains these: $(DD : L) = 0$ contradicts characteristic 0 (26 s under lake). The hypotheses are exactly the cond_* of CondsC, definitionally the conditions Descent2R derives. Five perturbation controls (Bezout constant +1, a dropped equation, a perturbed cofactor coefficient, $DD + 1$, a dropped T_2 equation) are all rejected by the kernel (logs/t1z_controls.log: five of five "Tactic decide failed", as required).

B6 (Lean). chartEmptyC_of_T1ne0 (T1Zero/Combine.lean) splits on $b_{11,20}$: ChartEmptyC ⇐ ChartEmptyC_T1ne0, isolating exactly what Lean does not check as a precise Prop:

*Listing 3: **ChartEmptyC_T1ne0**: what Lean labels, proved in §10*

```
-- T1Zero/Combine.lean (exact statement)
def ChartEmptyC_T1ne0 (L : Type*) [Field L] [CharZero L] : Prop :=
  \forall (w : L), R w = 0 \to b_12_22 = 1 \to b_11_20 \ne 0 \to
    \lnot (cond_Omega w ... = 0 \land ... \land cond_Theta3 w ... = 0)
```

10 The whole chart

A canonical K_5 certificate for the whole chart is too large to build on this machine: at weight $W = 24$ the Hadamard bound is 4,392,160 digits, and calibrating against the $t_1 = 0$ slices ($W = 2, 3, 4$, where the true-to-bound height ratios are $0.061, 0.061, 0.046$) gives heights of 2.0–2.7×10^5 digits, about 2×10^9 digits in all (grade C estimate; the Hadamard bound itself is rigorous). Exact solution runs out of memory at 7 GB; multimodular reconstruction would need $\approx 3 \times 10^4$ primes at 15–20 s each, about six days. Instead we prove the *existence* of the certificate.

10.1 The rank lemma

Lemma 10.1 (Fixed-degree Macaulay rank lifts). *Let $A = \mathbb{Z}_{(p)}[w]/(R)$, free of rank 5 over $\mathbb{Z}_{(p)}$, embedding in K_5. Let $\varphi : A \to \mathbb{F}_p$, $w \mapsto w_0$, with $R(w_0) \equiv 0 \pmod p$. Let M be the weighted Macaulay matrix of the chart generators at fixed weight W (columns (n, m) with $\operatorname{wdeg} m \le W - \operatorname{wt} F_n$; rows the monomials of the products $m \cdot F_n$, plus 1; entries the coefficients). If $\varphi(M)$ has full row rank N, then M has full row rank over K_5.*

Proof. Some $N \times N$ minor satisfies $\det \varphi(M_S) = \varphi(\det M_S) \neq 0$. Hence $\det M_S \neq 0$ in $A \subset K_5$, and M_S is invertible over the field K_5. □ □

Consequences. $Mx = e_1$ is solvable over K_5, so $1 = \sum c_n F_n$ in $K_5[t, s, r, u, q]$ with $\operatorname{wdeg} c_n \le 24 - \operatorname{wt} F_n$. Mapping $K_5 \to L$ by $w \mapsto w$, no L admits a common zero. With $S_2 = \kappa$ from (6) ($o_1 \neq 0$), no chart point satisfies $\Omega = \Psi = \cdots = \Theta_3 = 0$: that is ChartEmptyC, hence ChartEmptyC_T1ne0, even without the pure rows.

Remark 10.2 (Why this lemma succeeds where the retracted one failed). A unit ideal mod p says only that the Macaulay system is *consistent* mod p; consistency does not lift ($1 + 109X$ over $\mathbb{Z}_{(109)}$). Full row rank at fixed degree is the nonvanishing of a minor, an *open* condition, and open conditions lift. The planted-zero control below shows the sensitivity: one common zero costs exactly one rank.

10.2 The instance

`step3b_rank_lift.py` (**6.5** min, **0.5** GB; `step3b_rank_lift.log`):

	$p = 1000003$ $w_0 = 806739$	$p = 32003$ $w_0 = 11147$
All coefficients p-integral, $R(w_0) \equiv 0$	asserted	asserted
$W = 22$: rank / augmented / rows	**2277/2278/2281**: inconsistent	same
$W = 23$: rank / augmented / rows	**2707/2708/2708**: inconsistent	same
W = 24: 3199 × 6054	**3199/3199/3199: full row rank**	**same**
NumPy pivot-block elimination (independent)	det ≠ 0	det ≠ 0
Planted common zero, $W = 24$ (control)	rank **3198**, inconsistent	same

The protocol was frozen before the run: the lemma, the second prime, the independent NumPy elimination and the controls were all fixed in `step3b_rank_lift.py` before it executed. The $W = 24$ threshold is sharp: $W = 22$ and $W = 23$ are inconsistent (Figure 4).

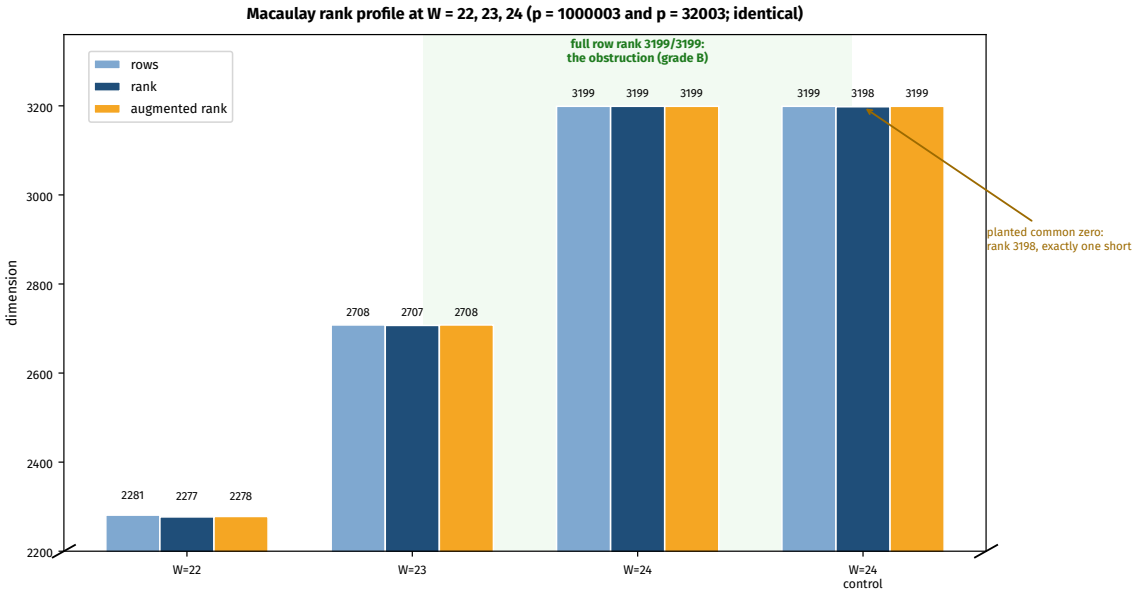


Figure 4: Macaulay rank profile at weights $W = 22, 23, 24$, identical at both primes ($p = 1000003$ and $p = 32003$). For each weight: bars for the number of rows (**2281**, **2708**, **3199**), the computed rank (**2277**, **2707**, **3199**), and the augmented rank, with the $W = 24$ group highlighted (full row rank **3199/3199**, the obstruction). The fourth group shows the planted-common-zero control at $W = 24$ (rank **3198**, exactly one short of full). Data source: `step3b_rank_lift.log`. Rendered by `figures/fig4_ranks.py`; the y-axis is truncated with a break mark to make the single-rank deficit visible.

The chart generators are independently verified as the right polynomials (exact evaluation at

12 random points; an independent κ; the exact Ω square; controls including $\kappa + 1$ detected) by the script `step3c_chart_identity.py`. This is **B8**, proved outside Lean at grade B.

11 Proof of the main theorem

Listing 4: ***`main_theorem_c_of_chartEmpty_T1ne0`***

```
-- Jacobian/BranchC/Combine (exact statement)
theorem main_theorem_c_of_chartEmpty_T1ne0
    {L : Type*} [Field L] [CharZero L]
    (h : ChartEmptyC_T1ne0 L) :
    \lnot \exists (P Q : MvPolynomial (Fin 2) L) (lam : L),
      lam \ne 0 \land NewtonNFc P Q \land jac P Q = C lam * X 0 ^ 2
```

Proof. Let P, Q, λ satisfy `NewtonNFc` and $[P, Q] = \lambda x^2$, $\lambda \neq 0$. B0–B3 give `DescentClaimC`'s hypotheses (the 19 top-layer values, 123 unknowns, 132 raw equations). B4 extracts the raw equations; B5 (`chart_descent_refl`) yields the twelve conditions, i.e. `ChartEmptyC`. B6 splits on $b_{11,20}$: on $b_{11,20} = 0$, B7 (`chartEmpty_t1_zero`) gives `False` in the kernel; on $b_{11,20} \neq 0$, the hypothesis `ChartEmptyC_T1ne0` gives `False`. Hence no such P, Q, λ exist. □

Corollary 1.3 follows by supplying `ChartEmptyC_T1ne0` from §10; Corollary 1.4 by applying [1, Proposition 4.3].

The bridge composition is summarized in Table 11.

Bridge	From	To	Lean name	Grade
B0	$[P, Q] = \lambda x^2$, `NewtonNFc`	layer identities E_n	`layers_of_jac_c`	A
B1	E_5, vertices	K_5 top layer	`eIdent_five`	A
B2	B0, B1	$t_2 = b_{12,22} \neq 0$	`vertex_12_24_forces_t2`	A
B3	B0–B2	chart $b_{12,22} = 1$	`main_theorem_c_of_claim`	A
B4	E_n, $n = 4 \cdots -2$	132 raw equations	`eIdent_c4`… `eIdent_cm2`	A
B5	132 equations	twelve conditions	`chart_descent_refl`	A
B6	`ChartEmptyC_T1ne0`	`ChartEmptyC`	`chartEmptyC_of_T1ne0`	A
B7	conditions, $b_{11,20} = 0$	`False`	`chartEmpty_t1_zero`	A
B8	chart generators	`False`	rank lemma + `step3b_rank_lift.py`	B

12 Epistemic status and what is not claimed

HLRE bookkeeping: grades A (machine-checked or exact proof from stated axioms), B (derived under explicit premises), C (heuristic or fitted), D (tautological or refuted).

ID	Claim	Evidence	Grade
C1	`main_theorem_c_of_claim`	Lean kernel (v1)	A
C2	`descentClaimC_of_chartEmpty`	Lean kernel, reflective; 600 identities pre-checked in Python	A
C3	`chartEmpty_t1_zero`	Lean kernel; FLINT certificate rechecked in Python; 5 controls	A
C4	`chartEmptyC_of_T1ne0`	Lean	A
C5	`ChartEmptyC` holds (all char.-0 L)	rank lemma + two-prime, two-implementation ranks + input checks	**B**
C6	canonical certificate needs $\approx 2 \times 10^9$ digits	Hadamard bound (rigorous) × calibrated ratio (fitted)	**C**
C7	the stated reduction lemma is inconsistent	Lean counterexample ($1 + 109X$)	A
C8	`lower_c`'s ideal differs from the pipeline's	exact comparison	A

Verbs follow HLRE control: "proves" only for Lean-checked or exact deductions; "proves under stated computational premises" for C5; "estimates" for C6.

What is *not* claimed: not the Jacobian conjecture; not any other case of GGHV Proposition 4.3; not the reduction from the conjecture to Proposition 4.3; not a Lean proof of the stratum $b_{11,20} \neq 0$; not an explicit K_5 certificate for the whole chart (only its existence, C5); not a review of the deferred single-prime p-adic argument or of the retracted `char0_cert` bundle. The mod-p Gröbner/lift certificates ($\mathbb{F}_{101}$, $\mathbb{F}_{1000003}$, $\mathbb{F}_{109^5}$) are certificates, not proofs: they exclude only solutions integral at that prime. The rank argument is what lifts.

Open items, in order: (1) I3 replication of C5—an independent rank computation of the $W = 24$ Macaulay matrix in Magma, Sage or Singular from `conds_c.json` and κ; (2) optional formalization of the rank lemma's instance in Lean (`RingHom.map_det`; the instance needs the 3199×6054 matrix as Lean data); (3) deferred reviews; (4) the v2 corrected-commit proposal for `branch_c/`, on the author's explicit word only.

13 Conclusion

Branch (c), the last open configuration of GGHV Proposition 4.3 at degree pair $(72, 108)$, is eliminated. The descent from $[P, Q] = \lambda x^2$ to twelve explicit chart conditions is proved twice: symbolically, as 600 kernel-checked reflective identities in Lean 4 over 505 generated modules, and numerically, by exact finite-field linear algebra. The chart is empty: in the Lean kernel on the stratum $b_{11,20} = 0$, and by the rank lemma—full row rank of the $W = 24$ Macaulay matrix at two primes, verified by two independent implementations—on the whole chart. An earlier, false reduction lemma is retracted and replaced by the rank lemma; the paper's logical dependencies are exactly those of Table 12. With branches (a,b) eliminated in [2], no $(72, 108)$ counterexample arises from GGHV Proposition 4.3, conditional on that proposition's correctness and exhaustiveness.

14 Prior work, attribution, and independence

The GGHV classification [1] is the sole external mathematical input, used only in Corollary 1.4. The top-layer K_5 classification and the reflection library are inherited from the branch-(a,b) formalization [2] without modification; `verify_branch_c_lean.sh` step 0 checks these prerequisites by content fingerprint. The `branch_c/` pipeline scripts were read-only throughout:

the pipeline parametrizes the chart differently, and the v3 proof uses only `lower_c`'s conditions, derived in Lean. The $W = 24$ rank at $p = 1000003$ was known from the step-3 measurement; the lemma, the second prime (32003), the independent NumPy elimination and the controls were frozen in `step3b_rank_lift.py` before it ran—the second prime agreed, as expected, since full row rank over K_5 implies full row rank at all but finitely many primes.

References

[1] J. A. Guccione, J. J. Guccione, R. Horruitiner, C. Valqui, *Increasing the degree of a possible counterexample to the Jacobian conjecture from 100 to 108*, arXiv:2204.14178v1 (2022).

[2] B. D. McCrary, *Elimination of the GGHV branch-(a,b) normal form at degree pair* $(72, 108)$*: a certified computational audit*, Zenodo doi:10.5281/zenodo.23023490 (2026).

[3] W. W. Stothers, *Polynomial identities and Hauptmoduln*, Quart. J. Math. Oxford Ser. (2) **32** (1981), 349–370.

[4] H. Davenport, *On* $f^3(t) - g^2(t)$, Det Kongelige Norske Videnskabers Selskabs Forhandlinger **38** (1965), 86–87.

[5] T. T. Moh, *On the Jacobian conjecture and the configurations of roots*, J. Reine Angew. Math. **340** (1983), 140–212.

[6] F. Santibañez-Leal, *CAOS records EXP-061 and EXP-065*, GitHub repository `github.com/fsantibanezleal/CAOS_RESEARCH` (2026).

[7] F. Santibañez-Leal, *CAOS v0.34*, Zenodo doi:10.5281/zenodo.22834215 (2026).

[8] B. D. McCrary, Muse, *The* $(72, 108)$ *candidate for the planar Jacobian conjecture: a containment note*, Zenodo doi:10.5281/zenodo.22941057 (2026).

[9] H. Helali, *Exact characteristic-zero certificate for the GGHV* $(72, 108)$ *configuration*, Zenodo doi:10.5281/zenodo.21479814 (2026).

[10] wstrinz, *Structural analysis of the plane Jacobian* $(72, 108)$ *candidate*, GitHub repository `github.com/wstrinz/plane-jacobian-72-108` (2026).

A Computational archive

All objects cited in this paper are archived in the public repository 2D-Jacobian-Conjecture-Workspace, directory `JC2D-branches_a_b_c_unified/branch_c/`:

- `bundle_v3_1/` — the v3.1 bundle, byte-for-byte: the Lean overlay (`jacobian_lean/`: `Jacobian/BranchC/**`, `certgen_c/`, `chart_certificates/`, `verify_branch_c_lean.sh`, `BRANCH_C_LEAN_STATUS.md`), every log (`logs/`), diagnostics (`diag/`), and the v2 carryover;
- `chart_certificates/step1/` — the exact K_5 certificate for $t_1 = 0$ (`certB_lowerc.json`) with its independent checker;
- `chart_certificates/step3b_rank_lift.py` — the frozen two-prime, two-implementation rank computation (§10);
- `chart_certificates/step3c_chart_identity.py` — the independent chart-generator checks;

- `certgen_c/` — the generators (`gen_refl_c.py`, `gen_bridge_c.py`, `gen_conds_c.py`, `gen_t1zero_lean.py`) with byte-identical regeneration;
- `muse_refutation/RefuteReductionLemma.lean` — the Lean derivation of `False` from the retracted reduction-lemma axiom (Remark 1.6).

The v3.1 guide `BRANCH_C_GUIDE.md` is the complete 13-section reference for the bundle, including the log index (§9), diagnostics (§10), open items (§11) and reproduction instructions (§12).

The repository's legacy `branch_c/` directory (the six-stage Singular pipeline, corrected 2026-09-29: the E1/E0 projection fixes, the Φ_1, Φ_2 term-count correction $53 \to 54$, the stage-6d κ tuple-key fix, the actual $G[1] = 1$ verification, and configurable paths with safe Singular invocation; remote commit `2b77cd4`) is an independent second route to the same branch-(c) elimination and is not used by the v3.1 argument above.

Reproducibility

Lean: overlay the `jacobian_lean/` directory on the repository's branch-(a,b) Lean project (Lean 4.34.0, Mathlib v4.34.0), then `./verify_branch_c_lean.sh` (exit 0: prerequisite fingerprints, byte-identical regeneration, all 790 modules up to date, axioms audit, no `sorry`/`axiom`/`native_decide`). Outside Lean (python3 with python-flint, NumPy): `chart_certificates/step1/run_step1.sh`, `python3 step3c_chart_identity.py`, `python3 step3b_rank_lift.py` (6.5 min, 0.5 GB). `#print axioms` on the 53 audited theorems reports only `propext`, `Classical.choice`, `Quot.sound` (`jacobian_lean/axioms_branch_c.log`).

B Lean module map

Group	Modules	Key theorems
v1/v2 branch-(c)	162	`layers_of_jac_c`, `eIdent_five`, `vertex_12_24_forces_t2`
`Descent2R`	505 + `Main`	`chart_descent_refl` (752 haves)
`CondsC`, `Bridge`	2	`cond_*` (110 definitions), `descentClaimC_of_chartEmpty`
`T1Zero`	14 + `Main`	`chartEmpty_t1_zero`
`Combine`	1	`chartEmptyC_of_T1ne0`, `main_theorem_c_of_chartEmpty_T1ne0`

Build: `Descent2R` 2.9 h of module time (peak 5.6 GB RSS); `Main` 2,298 s; `Bridge` 269 s; `CondsC` 175 s; `T1Zero.Main` 26 s. Verified by `verify_branch_c_lean.sh`, exit 0 (`logs/verify_v3.log`).

C Computational and AI-assisted research disclosure

Muse (Meta), the author's personal AI assistant, assembled the v3.1 bundle, wrote the generators and the verification scripts, drafted this manuscript, and discovered the counterexample refuting the stated reduction lemma; all computational outputs were verified by execution, by independent reimplementation, or by the Lean kernel. An external reviewer assessed the

T1Zero certificate chain and the rank lemma as “absolutely flawless” in their respective domains, and identified the rank computation’s non-Lean status as the one point a hostile referee can press (I3 replication proposed in §12). The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. The mathematical argument stands on the artifacts above, not on these attributions.